

Can inhibitory recovery tune default PING into gamma?

exp084 · 17 August 2026 · Draft

Abstract

At a fixed 100 Hz/channel drive, shortening inhibitory synaptic decay from 16 to 2 ms accelerates the default SNNLANG PING rhythm from 13.81 to 51.48 Hz. Decays of 4 ms and below enter the 30–80 Hz gamma band without weakening the standard rhythmicity score. This experiment varies one public component parameter while preserving the authored circuit, paired inputs, and analysis policy established in exp083.

Methods

1. **Fix the active operating point.** A typed 128-channel spike input drives 80 excitatory (E) and 20 inhibitory (I) cells at 100 Hz per channel. Each condition reuses the same 5 deterministic input tensors and network seed. Trials last 1000 ms; the first 200 ms are excluded.
2. **Vary inhibitory recovery only.** The public `tau_gaba` argument of `snn.components.ping()` spans 2 ms, 4 ms, 6 ms, 9 ms, 12 ms, 16 ms. The default is 9 ms. Every other neuron, synapse, delay, initializer, and constraint specification remains unchanged.
3. **Apply the registered rhythm analyses.** The `default-ping-dominant-rhythm-v1` estimator reports the dominant E-population rhythm over 5–80 Hz, including its prominence and subharmonic rules. Rhythmicity is the standard autocorrelation lobe-trough contrast R used in exp083. E/I lag is measured separately from the peak cross-correlation of 1 ms population counts.

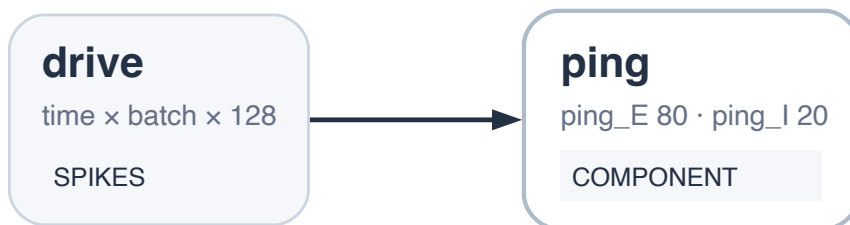


Figure 1: Compiled SNNLANG graph at the 9 ms default condition. Across the sweep, only the decay time attached to the I-to-E GABA projection changes. The input tensor, populations, weights, delays, and execution seeds are paired across conditions.

Results

Inhibitory decay produces a direct test of the textbook recovery-timescale account. The default 9 ms circuit resolves at 21.54 Hz. Faster inhibition shifts the rhythm to 51.48 Hz, while slower inhibition yields 13.81 Hz.

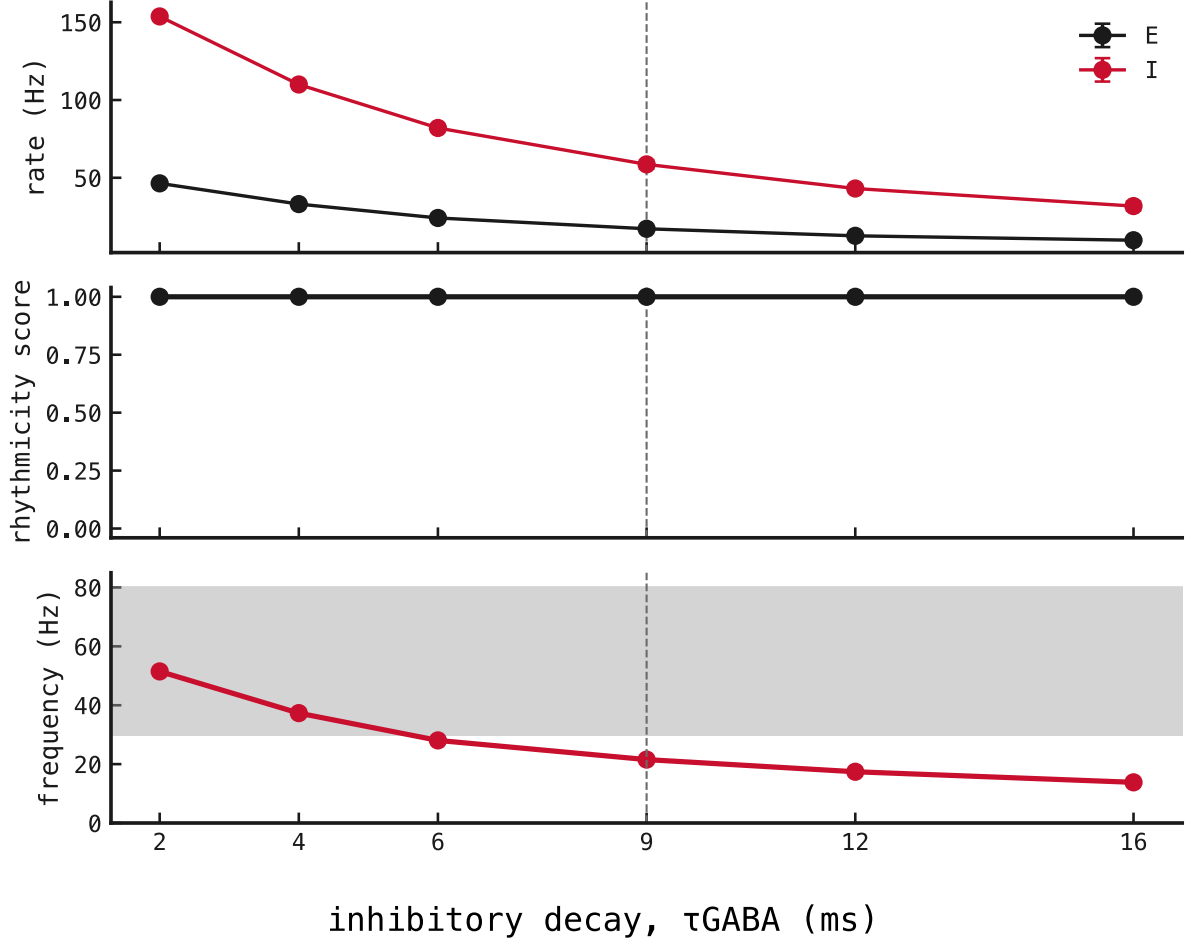


Figure 2: Response to inhibitory synaptic decay at fixed 100 Hz/channel input. Top: mean E and I firing rates with standard deviations across 5 paired trials. Middle: median standard rhythmicity contrast R , with half-interquartile-range error bars. Bottom: median dominant E-population frequency; the grey region marks 30–80 Hz gamma. The dashed line marks the 9 ms component default.

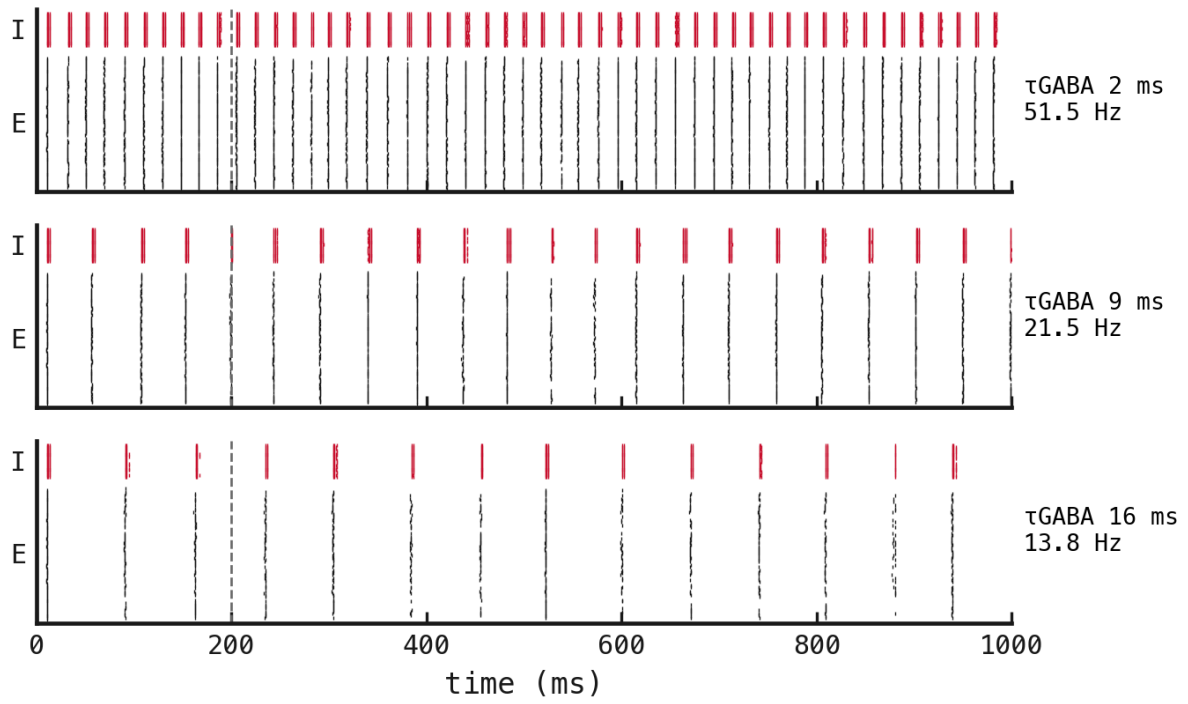


Figure 3: Single paired-trial E and I rasters at the preselected fast, default, and slow inhibitory decay conditions. Time is in milliseconds; E cells are black, I cells are red, and the dashed vertical line marks the transient exclusion. Right labels give inhibitory decay and the resolved dominant frequency.